

Enhancing severe weather predictions with the I-ConvGRU model: An iterative approach for radar echo time series through ConvGRU and RainNet integration

Rong Yang ^a, Hao Wang ^{a,b,c,*}, Fugui Zhang^{a,d}, Qiangyu Zeng^a, Taisong Xiong^a, Zhihao Liu^a and Hongfei Jin^a

^a College of Meteorological Observation, Chengdu University of Information Technology, Chengdu 610225, China

^b China Meteorological Administration Radar Meteorology Key Laboratory, Nanjing 21000, China

^c Wenjiang National Climatology Observatory, Sichuan Provincial Meteorological Service, Chengdu 611130, China

^d The Key Laboratory of Atmospheric Sounding, China Meteorological Administration, Chengdu 610225, China

*Corresponding author. E-mail: wh@cuit.edu.cn

 RY, 0009-0001-3694-0059; HW, 0000-0003-0090-2840

ABSTRACT

Precipitation nowcasting plays a crucial role in disaster prevention and mitigation. Existing forecasting models often underutilize output data, leading to suboptimal forecasting performance. To tackle this issue, we introduce the I-ConvGRU model, a novel radar echo timing prediction model that synergizes the temporal dynamics optimization of ConvGRU with the spatial feature enhancement capabilities of RainNet. The model forecasts future scenarios by processing 10 sequential time-series images as input while employing skip connections to boost its spatial feature representation further. Evaluation of the radar echo data set from the Hong Kong Hydrological and Meteorological Bureau spanning from 2009 to 2015 demonstrates the I-ConvGRU model's superiority, with reductions of 17(3.8%) and 49(3.2%) in MSE and MAE metrics, respectively, compared with the TrajGRU model; meanwhile, the I-ConvGRU model had 52(5.8%) and 144(3.8%) lower values on the B-MSE and B-MAE metrics, respectively, than the slightly better performing TrajGRU model. Notably, it significantly improves the prediction of severe precipitation events, with the CSI and HSS metrics increasing by 0.0251(9.6%) and 0.0277(6.8%). These results affirm the model's enhanced effectiveness in radar echo forecasting, particularly in predicting heavy rainfall events.

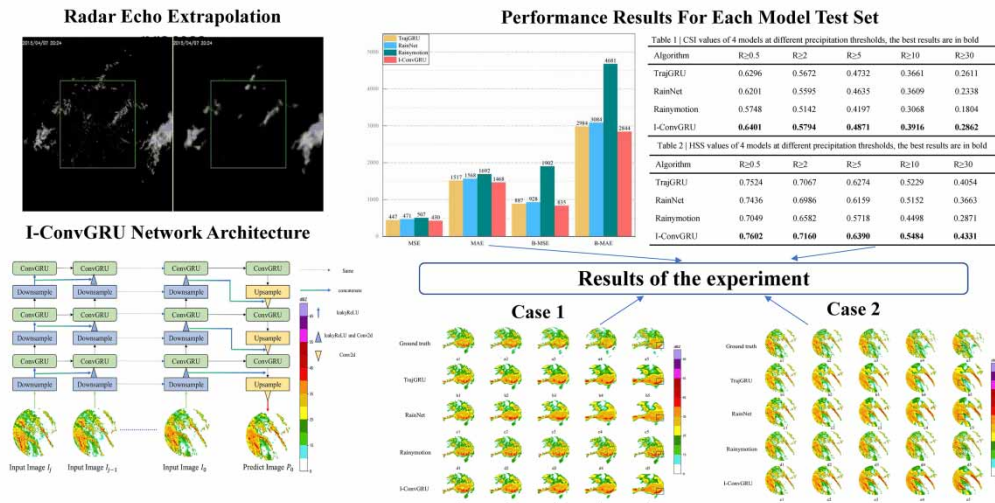
Key words: heavy rainfall, nowcasting, radar echo, time prediction

HIGHLIGHTS

- The model designed by combining recurrent neural networks and iterative networks improves the warning rate of radar echo extrapolation.
- The model has strong early warning ability for heavy rainfall.

GRAPHICAL ABSTRACT

Enhancing Severe Weather Predictions with the I-ConvGRU Model: An Iterative Approach for Radar Echo Time Series Through ConvGRU and RainNet Integration



INTRODUCTION

Severe convective weather is characterized by its limited spatial extent, brief duration, intense unpredictability, and significant destructive potential (Wang *et al.* 2022; Chowdhury *et al.* 2023; Xiang *et al.* 2023). It often triggers catastrophic conditions, including thunderstorms, hail, strong winds, and heavy rainfall (Zhang & Melhauser 2012). Additionally, it represents a significant natural hazard in China (Zhang & Melhauser 2012) and it is also a key challenging aspect in current meteorological forecasting operations (Yan *et al.* 2022). While conventional numerical meteorological models, enhanced by high-performance computing and data assimilation techniques, have markedly improved weather forecasting accuracy (Bauer *et al.* 2015), the computational demands for predicting precipitation dynamics at a high spatiotemporal resolution are generally not well suited for real-time nowcasting operations, which require frequent updates (every 5–10 min). Furthermore, due to their inherently nonlinear characteristics, traditional dynamic framework-based prediction methods struggle to forecast severe convective weather accurately.

Forecasting severe convective weather is typically categorized into short-term forecasts and nowcasts. A short-term forecast covers 0–12 h, focusing primarily on the potential for extreme weather events within this timeframe. Conversely, nowcasting, which spans a shorter lead time of up to 2 h (Shukla *et al.* 2014), emphasizes immediate responses to imminent weather changes. Weather radar serves as a pivotal tool in nowcasting, offering an unparalleled spatial and temporal resolution for tracking precipitation systems and forecasting medium to small-scale severe convective events (Wang *et al.* 2023). Currently, radar echo extrapolation is the principal technique in nowcasting. This method extrapolates the future location and intensity of radar echoes based on current weather radar observations to predict severe convective weather (Rinehart & Garvey 1978; Chen *et al.* 2022). Many studies have confirmed that heuristic extrapolation of precipitation dynamics, as observed by weather radar, consistently surpasses the accuracy of numerical forecast models for short lead times (Lin *et al.* 2005; Yin *et al.* 2021). With the ongoing expansion and operation of weather radar networks, advancements in radar echo extrapolation algorithms leveraging radar observation data are crucial for minimizing the impact of meteorological disasters (Shi *et al.* 2018).

Current research in echo extrapolation predominantly utilizes the single-body centroid, cross-correlation, and optical flow methods. The single-body centroid method, rooted in three-dimensional thunderstorm tracking technology (Sun *et al.* 2014), identifies and analyzes single-body features within the echo to predict the echo's future position. This approach has proven effective for tracking monomers with strong echoes and compact volumes, and various algorithms based on this method have seen widespread application (Dixon & Wiener 1993; Johnson *et al.* 1998; Dai *et al.* 2022). The cross-correlation method, or tracking radar echoes by correlation (TREC) (Zou *et al.* 2019), leverages optimal spatial correlations of echoes to trace precipitation system movements. It accounts for echo size, movement direction, and deformations occurring during transit, achieving notable success in nowcasting operations (Li *et al.* 1995, 2013). However, each method has its constraints: the

single-body centroid method is best suited for convective severe thunderstorm cells and less effective in general convective weather forecasting. The cross-correlation method excels in slowly changing stratiform cloud precipitation systems but falters with rapidly changing precipitation echoes, as rapid weather changes introduce significant errors in motion vector field calculations, diminishing forecast accuracy (Zhu & Dai 2022). The optical flow method, extensively applied in practice, originates from computer vision. A notable variant is the variational optical flow method, ROVER (Woo & Wong 2017), which calculates the optical flow field of consecutive radar images based on a stationary assumption, employing the technique suggested by Brox *et al.* (2004) and implementing the semi-Lagrangian advection scheme for precipitation forecasting (Tishchenko *et al.* 2019). Although the optical flow method has an inherent physical change mechanism, the separation between calculation and extrapolation complicates the determination of model parameters, presenting challenges in its application.

In recent years, the advent of large-scale parallel computing and the widespread adoption of graphics processing unit (GPU) devices have significantly enhanced computing capabilities. Following the establishment of a new generation of weather radar networks in China (Min *et al.* 2019), there has been an exponential increase in the volume of data available for training models. Consequently, deep learning has increasingly been applied to radar echo extrapolation. With its robust feature recognition capabilities, deep learning can discern complex patterns and extract meaningful insights from radar echo data, leading to more accurate extrapolation predictions. Additionally, deep learning models excel in modeling complex non-linear relationships inherent in radar echo data, enhancing forecasts' precision and reliability (Zhang *et al.* 2022). The most notable deep learning models recently deployed in this field fall into three main categories: recurrent neural networks (RNNs), convolutional neural networks (CNNs), and generative adversarial networks (GANs).

In the realm of radar echo extrapolation, RNNs have seen significant application, with long short-term memory networks (LSTMs) and gated recurrent units (GRUs) being particularly prominent. Shi *et al.* (2015) introduced the ConvLSTM model, which integrates a convolution layer and a sampling layer to extract spatial characteristics of sequences and accurately capture the spatiotemporal dynamics of radar echo patterns. However, due to its complexity, Shi *et al.* (2017) later proposed more streamlined architectures, namely the Trajectory Gated Recurrent Unit (TrajGRU) and Convolution Gated Recurrent Unit (ConvGRU). The TrajGRU model, which accounts for motion trajectories, achieves superior results compared with ConvLSTM, prompting extensive research and development on the ConvGRU and ConvLSTM networks by meteorologists.

Yuan *et al.* (2018) proposed the model of adaptive ALO-LSTM by combining with ALO module on the basis of LSTM network. It has a better ability to adapt to the data. However, it does not address the problem that the ability of the model to capture long time dependencies still diminishes as the sequence grows. To address this problem, Jing *et al.* (2020) proposed a hierarchical prediction recurrent neural network (HPRNN) to solve the problem of increasing prediction error over time. MotionGRU (Wu *et al.* 2021) units were proposed for GRU networks to capture more complex spatiotemporal motions. However, their structures are complex compared with the ConvGRU model. The fusion for iterative networks needs to waste more computational resources.

Additionally, CNNs have been explored for their ability to extract image features efficiently and associate them with label images (Sharif Razavian *et al.* 2014). A standout model in this category is U-Net (Agrawal *et al.* 2019), which employs a fully convolutional structure with upsampling and downsampling layers, gaining acclaim for its spatial feature extraction prowess and serving as a foundational network for full CNN models. Nie *et al.* (2021) proposed SmaAt-UNet on the basis of U-Net network model, which introduces the channel attention module in the model. The feature extraction ability of the model is enhanced. And due to the simplicity of the structure of the U-Net model, meteorologists also used it in multi-source data fusion prediction. The RainNet model, introduced by Ayzel *et al.* (2020), builds on U-Net by incorporating a recursive network and demonstrates exceptional performance in nowcasting precipitation. Pan *et al.* (2021) proposed the FURENet model with U-Net as the backbone network, which satisfies the purpose of multivariate information input. Polarized radar parameters specific differential phase (KDP) and differential reflectivity (ZDR) are input into the model to improve the accuracy of precipitation proximity forecasting.

GANs have also been merged with RNN and CNN frameworks to enhance early warning accuracy. Tian *et al.* (2020) developed the GA-ConvGRU model, incorporating a GAN with the ConvGRU model and outperforming both the ConvGRU and optical flow methods. Wang *et al.* (2021) proposed an advanced model, ExtGAN, based on a conditional GAN, which surpasses the optical flow method and 3D-CNN models in effectiveness. Notably, the deep generative model of radar (DGMR) by Ravuri *et al.* (2021) stands out as a seminal contribution, significantly improving the resolution of predicted radar images and consistently outperforming both the U-Net and PySTEPS (Pulkkinen *et al.* 2019) models in most forecasting scenarios.

And due to the wide application of MSE, MAE loss function weighting in traditional methods. It causes blurring of radar images. So many meteorologists use GAN to improve the clarity of radar image. Niu *et al.* (2024) introduced conditional generative adversarial network (CGAN) network on the basis of encoder-predictor network greatly improved the clarity of the model.

In summary, while the RainNet neural network exhibits certain limitations, conventional models like Rainymotion, based on optical flow (Ayzel *et al.* 2019), struggle to deliver accurate predictions under heavy precipitation conditions. However, the recursive method utilized within the RainNet model shows promise. In light of this, our study proposes integrating the RainNet network with the ConvGRU network to develop a novel neural network model, the I-ConvGRU (iteration convolutional gated recurrent unit model). This integration leverages the recursive method to shorten the prediction horizon and selects the ConvGRU model for its simplicity to enhance temporal feature extraction within the I-ConvGRU model. This research work offers several advantages over current mainstream methodologies: Firstly, unlike the traditional encoder-predictor, the I-ConvGRU model has a simpler structure, which does not increase the complexity of the model due to the increase of the output signal. In addition, the ConvGRU model is selected as the basic structure, and its spatiotemporal feature extraction ability is obviously higher than that of the convolutional network, and the ConvGRU network structure is relatively simple, saving some computing resources. At the same time, putting the output signal back into the network will introduce more temporal and spatial features, and improve the ability of the network to forecast heavy precipitation. Finally, some jump connections are introduced in the sampling process to help the network introduce multi-scale information. It is helpful to improve the ability of the model to extract features.

The second section of this study details the physical model, experimental data, and evaluation methods employed in deep learning. The third section presents the model's testing results and discussions. The concluding section summarizes the findings of this paper.

MODELS DATA AND METHODS

Model introduction

Our study introduced the I-ConvGRU model, a novel recurrent neural network that represents a fusion of ConvGRU and RainNet technologies (Figure 1). This model diverges from conventional encoder-predictor frameworks by adopting a

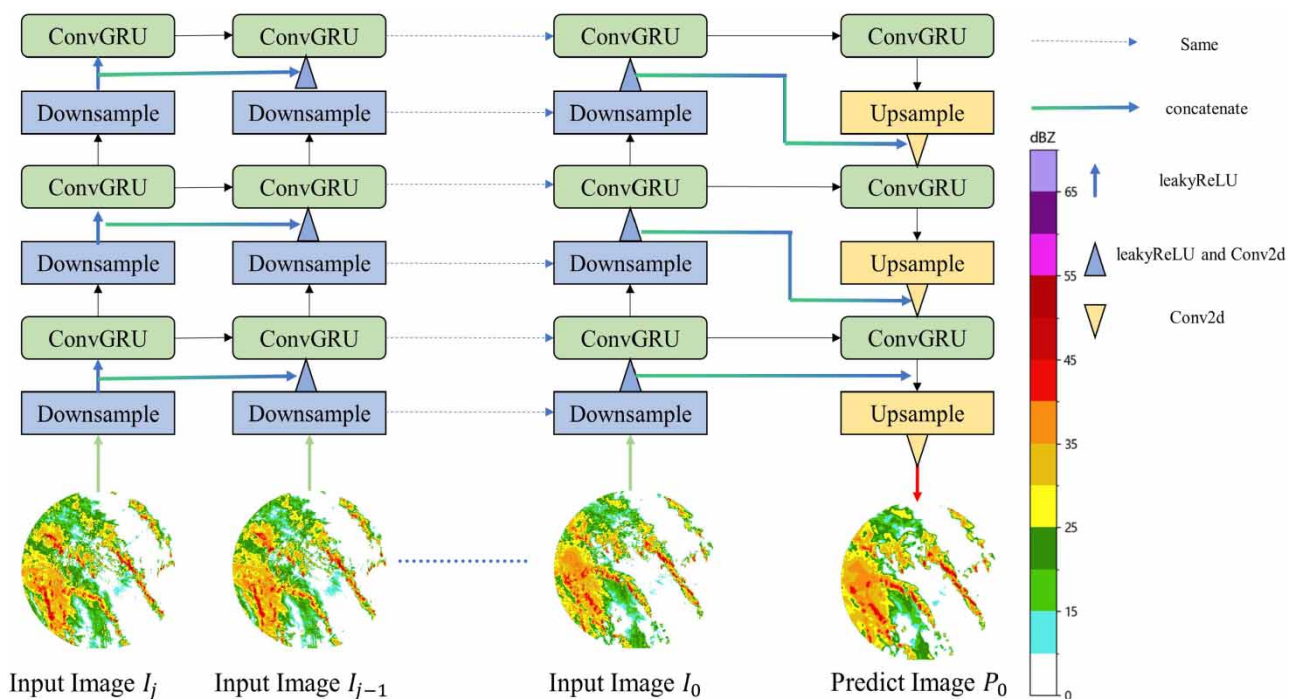


Figure 1 | I-ConvGRU model structure diagram.

unique approach to input and prediction. Specifically, the model forecasts a single subsequent image for each input sequence of images, denoted by j . This predicted image is then recycled back into the input data set, taking the position of the most recent input image. This process is iteratively repeated, allowing for sequential output generation.

The I-ConvGRU model merges the strengths of traditional encoder-predictor networks with a novel iterative approach. The encoder achieves spatial reduction via downsampling layers, and convolution extracts image spatial features. The decoder then utilizes upsampling layer deconvolution to enlarge image feature dimensions, progressively enhancing resolution. The model is structured with 30 downsampling layers, 33 ConvGRU networks, 30 LeakyReLU activation functions, 33 convolution operations, and 3 upsampling layers. A crucial addition is a 1×1 convolution layer post-tensor splicing aimed at channel compression and feature fusion. The final output stage employs multi-layer convolution to extract spatial features from input, driven by data. The model uses the LeakyReLU activation function prior to linking the downsampling layer with the ConvGRU network, incorporating a minor slope in its negative part to prevent gradient vanishing issues, thereby boosting training stability and effectiveness, as well as enhancing the model's nonlinear fitting capacity (Jiang & Cheng 2019).

This article models based on convolution (ConvGRU), and its basic formula is as follows (Shi *et al.* 2017):

$$Z_t = \sigma(W_{xz} * X_t + W_{hz} * H_{t-1}) \quad (1)$$

$$R_t = \sigma(W_{xr} * X_t + W_{hr} * H_{t-1}) \quad (2)$$

$$H'_t = f(W_{xh} * X_t + R_t \circ (W_{hh} * H_{t-1})) \quad (3)$$

$$H_t = (1 - z_t) \circ H'_t + Z_t \circ H_{t-1} \quad (4)$$

For simplicity, the deviation part is omitted from the formula, $*$ indicating the convolution operation; $\circ$ indicating the Hadamard product. H_t , R_t , Z_t and H'_t represent the memory state, reset gate, update gate, and information, respectively (Figure 2); X_t is the input; f is the activation function, which is ReLU with leakage correction with a negative slope of 0.2. Whenever new input arrives, the reset gate controls whether the previous state is cleared, and the update gate controls the amount of new information entering the state.

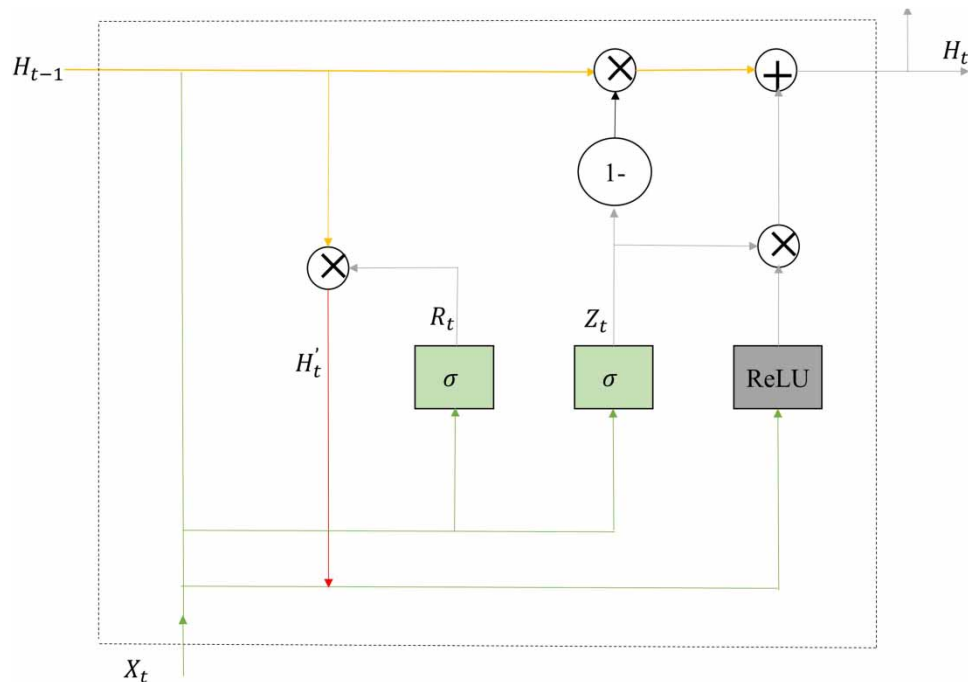


Figure 2 | The structure of a single GRU neuron in the ConvGRU model.

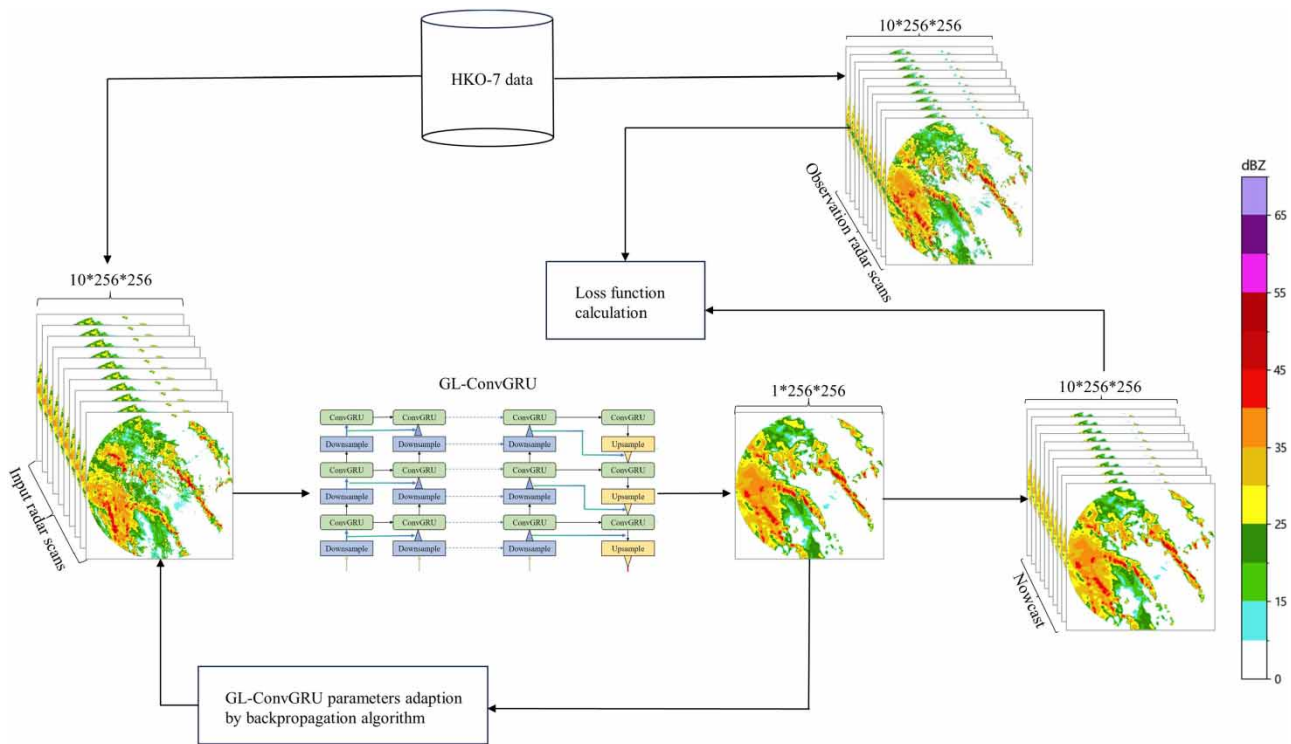


Figure 3 | Technical route for I-ConvGRU model echo extrapolation.

Figure 3 outlines the technical methodology of our study. Initially, we randomly selected 20 consecutive images from the HKO-7 data set for training, splitting them into two segments: 10 images served as observed data (true value), and the other 10 served as input data. These input images are processed through the network to predict the radar image for $t + 6$ min. This predicted image is then incorporated back into the input data set, replacing the oldest image ($t - 60$), and the process is repeated to sequentially generate predictions up to $t + 12$ min for 10 predicted images. This iterative recycling of predicted images as new inputs allows for incorporating additional spatiotemporal information, thereby enhancing model optimization. The model parameters of the I-ConvGRU are subsequently refined using the backpropagation algorithm to minimize the loss function, ensuring precise and reliable forecasts.

Test data

In this study, we will use the HKO-7 data set collected by the Hong Kong Hydrological and Meteorological Bureau as data to train and verify the early warning capabilities of the I-ConvGRU model. In order to achieve better prediction results, the data set has undergone denoising preprocessing, and the effectiveness of the data set has been verified on the ConvGRU and Traj-GRU models. He *et al.* (2022) and Zeng *et al.* (2022) used this data set to verify the model's performance. The HKO-7 data set contains Doppler weather radar observation data in Hong Kong from 2009 to 2015. What is stored is the radar echo CAPPI map of the contour plane after interpolating the reflectivity. The time resolution is 6 min. In order to reduce the dBZ of the image itself, we convert the image reflectivity factor to a grey scale image taking values from 0 to 255.

$$pixel = 255 * \frac{dBZ + 10}{70} + 0.5 \quad (5)$$

This formula *pixel* represents the pixel value; and *dBZ* represents the reflectivity. The radar echo image is based on the conversion of reflectivity. This formula shows that the larger the pixel value, the greater the reflectivity, the stronger the radar echo intensity, and the greater the rainfall. The images used in the data set are 480*480 resolution images, including a 512*512 km range centered on Hong Kong. To speed up the calculation of the model, we reduce the image size to 256*256 resolution.

We divided the HKO-7 data set into three parts: training set, verification set, and test set. The period from 2009 to 2014 is divided into training and validation sets. The number of days in the training set is 812 days, and the number of days in the validation set is 50 days. The test set selects data from 2015, totaling 131 days. During the model training process, we set the batch size `batch_size` to 2, the learning rate of Adam to 1×10^{-4} , the learning strategy to ReduceLRPlateau, iterations 100,000 times, and the model is saved every 5,000 times, which is consistent with the data of the validation set. Compare and select the best model and then test it. All models are implemented on pytorch and run on NVIDIA Tesla P40 graphics card with 24 GB of video memory.

During the model training process, we convert reflectance into precipitation through the formula:

$$dBZ = 10 \log a + 10b \log R \quad (6)$$

where a is equal to 58.53 and b is equal to 1.56. dBZ represents reflectivity, and R represents precipitation. However, due to the uneven rainfall distribution in the HKO-7 data set (Figure 4), the part with rainfall $x \geq 30$ mm/h only accounts for 0.42% of the total.

To balance the rainfall and improve the prediction rate, we set corresponding weights for each part of the rainfall. This paper selects the weights used in the TrajGRU model.

$$\omega(x) = \begin{cases} 1, & x < 2 \\ 2, & 2 \leq x < 5 \\ 5, & 5 \leq x < 10 \\ 10, & 10 \leq x < 30 \\ 30, & x \geq 30 \end{cases} \quad (7)$$

Evaluation methods

Common evaluation indicators for regression problems include mean absolute error (MAE) and root mean squared error (MSE), where MAE is the average of the absolute errors between the statistical prediction value and the true value, and MSE is the average sum of the squared errors of the statistical predicted value and the true value. The calculation formulas of MAE and MSE are as follows:

$$MAE = \frac{1}{N} \sum_{n=1}^N \sum_{i=1}^{480} \sum_{j=1}^{480} |x_{n,i,j} - \hat{x}_{n,i,j}| \quad (8)$$

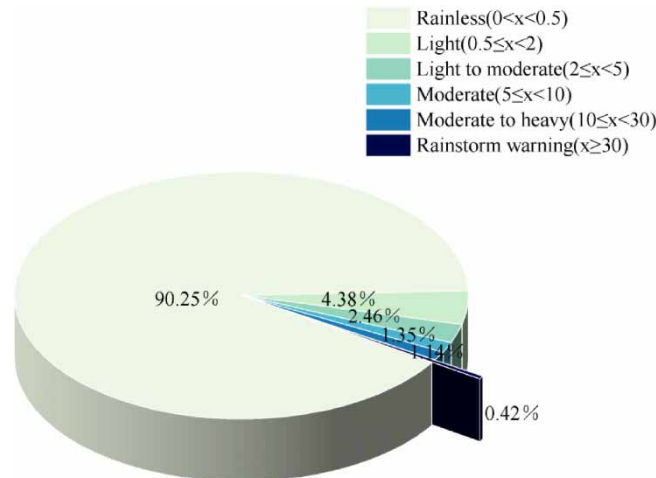


Figure 4 | HKO-7 data set rainfall distribution map.

$$\text{MSE} = \frac{1}{N} \sum_{n=1}^N \sum_{i=1}^{480} \sum_{j=1}^{480} (x_{n,ij} - \hat{x}_{n,ij})^2 \quad (9)$$

In the formula, N is the total number of inputs, $x_{n,ij}$ represents the true value of the n th picture at the position of the radar echo image (i, j) , and $\hat{x}_{n,ij}$ represents the size of the predicted value at the same position. Since the rainfall distribution in this paper is uneven, we propose balanced mean absolute error (B-MAE) and balanced mean squared error (B-MSE) as evaluation indicators. Using their sum(1:1) as the loss function (LOSS) improves the prediction performance. The calculation formulas of B-MAE, B-MSE, and LOSS are as follows (Zeng *et al.* 2022):

$$\text{B-MAE} = \frac{1}{N} \sum_{n=1}^N \sum_{i=1}^{480} \sum_{j=1}^{480} \omega_{n,ij} |x_{n,ij} - \hat{x}_{n,ij}| \quad (10)$$

$$\text{B-MSE} = \frac{1}{N} \sum_{n=1}^N \sum_{i=1}^{480} \sum_{j=1}^{480} \omega_{n,ij} (x_{n,ij} - \hat{x}_{n,ij})^2 \quad (11)$$

$$\text{LOSS} = \frac{1}{N} \sum_{n=1}^N \sum_{i=1}^{480} \sum_{j=1}^{480} \omega_{n,ij} (x_{n,ij} - \hat{x}_{n,ij})^2 + \frac{1}{N} \sum_{n=1}^N \sum_{i=1}^{480} \sum_{j=1}^{480} \omega_{n,ij} |x_{n,ij} - \hat{x}_{n,ij}| \quad (12)$$

$\omega_{n,ij}$ is the weight of the n th picture at the position of the radar echo map (i, j) . Different weights will be assigned to different positions. The stronger the intensity, the greater the weight to balance the precipitation. This will increase the model's attention to rainfall prediction and better predict heavy rainfall.

Due to different precipitation thresholds, different attention will be paid to meteorological operations. Therefore, evaluating the prediction performance of models under different thresholds is of great significance for echo extrapolation early warning. To assess the prediction performance of the model under different thresholds, we used the critical success index (CSI), false alarm rate (FAR), and Heidke skill score (HSS) for detailed evaluation. The calculation formulas of CSI, FAR, and HSS are as follows:

$$\text{CSI} = \frac{\text{TP}}{\text{TP} + \text{FN} + \text{FP}} \quad (13)$$

$$\text{FAR} = \frac{\text{FP}}{\text{TP} + \text{FP}} \quad (14)$$

$$\text{HSS} = \frac{\text{TP} \times \text{TN} - \text{FN} \times \text{FP}}{(\text{TP} + \text{FN})(\text{FN} + \text{TN}) + (\text{TP} + \text{FP})(\text{FP} + \text{TN})} \quad (15)$$

We treat ground observations as true values, and the result is 1 when the radar echo is higher than the threshold set by the data set and 0 otherwise. In the above formula, True Positive (TP) means (prediction = 1, true value = 1), False Negative (FN) means (prediction = 0, true value = 1), and False Positive (FP) means (prediction = 1, true value = 0) and True Negative (TN) representation (prediction = 0, true value = 0). The higher the CSI and HSS values, the better the prediction effect, and the lower the FAR value, the better the prediction effect.

Meanwhile, in order to study the radar warning pictures in a more refined way, we introduce the structural similarity index (SSIM), and peak signal-to-noise ratio (PSNR), which are indicators to evaluate the quality of the radar warning pictures: the specific formulas of SSIM and PSNR are as follows:

$$\text{SSIM}(x, y) = \frac{(2\mu_x\mu_y + c_1)(2\sigma_{xy} + c_2)}{(\mu_x^2 + \mu_y^2 + c_1)(\sigma_x^2 + \sigma_y^2 + c_2)} \quad (16)$$

$$\text{PSNR} = 10 \log_{10} \left(\frac{\text{MAX}^2}{\text{MSE}} \right) \quad (17)$$

The SSIM metrics x and y represent two images, respectively; μ_x and μ_y represent the luminance averages of the two images; σ_x^2 and σ_y^2 represent the luminance variance of the two images; σ_{xy} is the luminance covariance of the two images; c_1 and c_2 are the stability parameters. MAX in PSNR denotes the maximum of all values and MSE denotes the root mean square error.

RESULTS AND DISCUSSION

To evaluate the effectiveness of the I-ConvGRU model, we compared it with three other models to assess its performance. The first comparison model is the TrajGRU, an RNN-based model that enhances the GRU network with motion trajectory capabilities, showing improved prediction accuracy over the traditional ConvGRU model. The second model, RainNet, leverages a CNN framework and introduces an iterative network built on the U-Net architecture, allowing for a comparative analysis of early warning capabilities across different models of iterative networks. The third model, Rainymotion, utilizes the optical flow method for precipitation field tracking and adopts a constant vector advection scheme for extrapolation. Rainymotion, part of the rainymotion library, demonstrates performance on par with, or even superior to, the RADVOR (Radar Real-time Forecast) model, covering a broad spectrum of rainfall events. To objectively validate the reliability and generalizability of the I-ConvGRU model, we selected the iteration of each model that yielded the lowest loss function for evaluation on the test set. We employed a comprehensive set of metrics, including MSE, MAE, B-MSE, B-MAE, CSI, HSS, FAR, etc., to conduct a thorough performance analysis of the I-ConvGRU model. The overall performance metrics of MSE, MAE, B-MSE, and B-MAE on the test set are detailed in Figure 5.

Data analysis reveals that the I-ConvGRU model outperforms other models in terms of MSE, MAE, B-MSE, and B-MAE for predictions within a 256*256 pixel area, indicating lower discrepancies between predicted images and actual observations. Specifically, compared with the TrajGRU model, which had the next best performance, the I-ConvGRU model's MSE and MAE metrics show reductions of 3.8 and 3.2%, respectively, with B-MSE and B-MAE also decreasing by 5.8 and 3.8%. This proves the superior performance of I-ConvGRU on the test set.

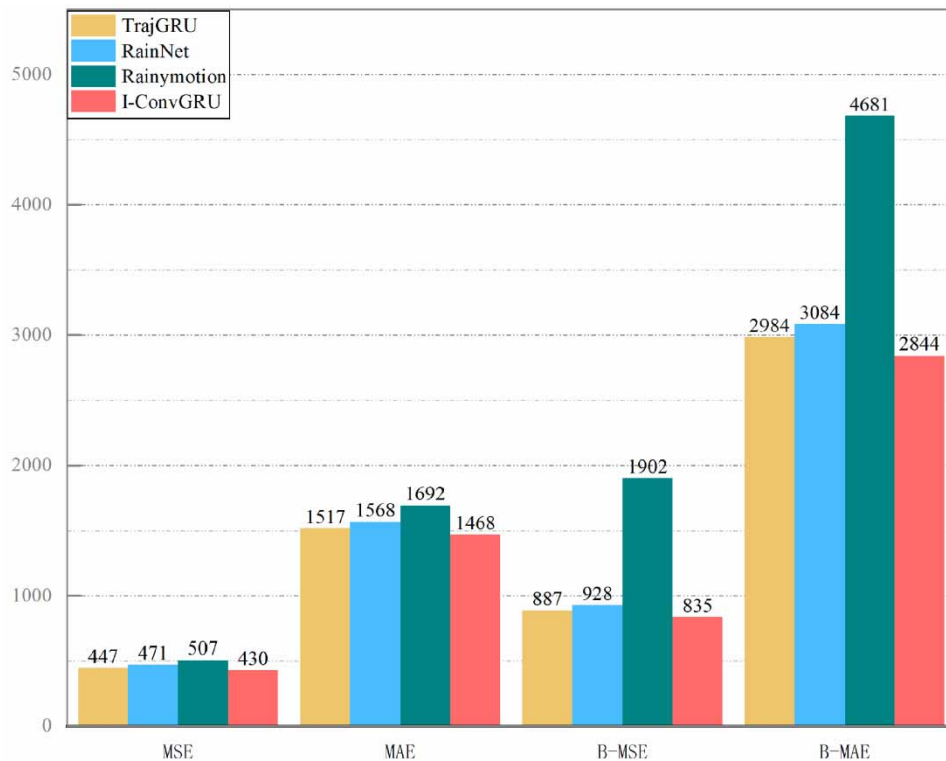


Figure 5 | Performance of four models on the test set of MSE, MAE, B-MSE, and B-MAE.

Table 1 | SSIM and PSNR values of model predictions on the test set for each model

Models	SSIM	PSNR
TrajGRU	0.7874	21.607
RainNet	0.7899	21.432
Rainymotion	0.7481	21.076
I-ConvGRU	0.7956	21.77

To more intuitively analyze the ability of the test set model to predict pictures. In Table 1, we give the PSNR and SSIM mean values of the pictures predicted by different models on the test set. It can be seen that Rainymotion has the worst structural similarity, followed by the TrajGRU model. The I-ConvGRU model has an SSIM value of 0.7956, which is the closest to 1. Analyzing the PSNRs of the models, it can be seen that the I-ConvGRU model has a PSNR value of 21.77, which is the highest among all the models. This indicates that the I-ConvGRU model has the strongest predictive ability.

For a more detailed assessment of each model's early warning capabilities, we examined the performance of CSI, FAR, and HSS across different thresholds (as detailed in Tables 2–4). The effectiveness of CSI and HSS diminishes as the threshold increases. However, the I-ConvGRU model consistently outperforms the comparative models across these metrics, demonstrating greater CSI and HSS values. Notably, at a precipitation threshold of ≥ 30 mm/h, the I-ConvGRU model's CSI and HSS metrics are increased by 9.6 and 6.8%, respectively, compared with the TrajGRU model. Although its FAR values are less favorable than those of the Rainymotion model at lower precipitation thresholds ($R \geq 2$ mm/h, $R \geq 5$ mm/h, and $R \geq 10$ mm/h), this is likely due to the optical flow method's pixel-level motion analysis, which more accurately captures

Table 2 | CSI values of four models at different precipitation thresholds, the best results are in bold

Algorithm	$R \geq 0.5$	$R \geq 2$	$R \geq 5$	$R \geq 10$	$R \geq 30$
TrajGRU	0.6296	0.5672	0.4732	0.3661	0.2611
RainNet	0.6201	0.5595	0.4635	0.3609	0.2338
Rainymotion	0.5748	0.5142	0.4197	0.3068	0.1804
I-ConvGRU	0.6401	0.5794	0.4871	0.3916	0.2862

Table 3 | FAR values of four models at different precipitation thresholds, the best results are in bold

Algorithm	$R \geq 0.5$	$R \geq 2$	$R \geq 5$	$R \geq 10$	$R \geq 30$
TrajGRU	0.2514	0.3548	0.4609	0.5541	0.6081
RainNet	0.2549	0.3601	0.4741	0.5662	0.6213
Rainymotion	0.2554	0.3153	0.4089	0.5280	0.6806
I-ConvGRU	0.2437	0.3489	0.4557	0.5424	0.5986

Table 4 | HSS values of four models at different precipitation thresholds, the best results are in bold

Algorithm	$R \geq 0.5$	$R \geq 2$	$R \geq 5$	$R \geq 10$	$R \geq 30$
TrajGRU	0.7524	0.7067	0.6274	0.5229	0.4054
RainNet	0.7436	0.6986	0.6159	0.5152	0.3663
Rainymotion	0.7049	0.6582	0.5718	0.4498	0.2871
I-ConvGRU	0.7602	0.7160	0.6390	0.5484	0.4331

motion trajectories and reduces false alarm rates. Nonetheless, the I-ConvGRU model shows superior performance across all metrics when the precipitation threshold is set at $R \geq 0.5$ mm/h and $R \geq 30$ mm/h, which is particularly notable during heavy rainfall events.

Overall, the I-ConvGRU model performs best on MSE, MAE, B-MSE, and B-MAE on the test set. The SSIM and PSNR values predicted by the I-ConvGRU model are also the largest. The pictures that proved their predictions were the most similar to the real ones. At the same time, the model has the best performance compared with other models when the threshold is above 30 mm/h. This is because the traditional method does not introduce the output information, so the prediction performance of the model is not as good as that of the I-ConvGRU method for a small range of heavy rainfall regions. At the same time, the addition of output information provides more features to the model. So its evaluation indicators are in the test set of the best performance.

In our comparative evaluation, we utilized sequences of 10 consecutive radar echoes to forecast the radar echo sequence for the next 10 time steps, effectively predicting 1 h of future data based on the previous hour's historical data. This approach aims to provide a clear and intuitive understanding of the prediction performance across different models. To accurately assess the models' capability to capture the dynamics of severe convective weather, we focused on areas characterized by strong echoes, a common indicator of such weather. Our analysis included cases that span the entire lifecycle of convective processes – namely, the genesis, development, and dissipation stages. This comprehensive approach allows us to evaluate the models' effectiveness in predicting severe convective weather events, highlighting their strengths and limitations in various stages of convective activity.

Case 1

This example selects an extreme short-term precipitation event in Hong Kong between 16:00 and 18:00 on May 23, 2015. The rainfall on that day reached 169.4 mm. As can be seen from Figure 6, we have given the radar echo images of five time periods

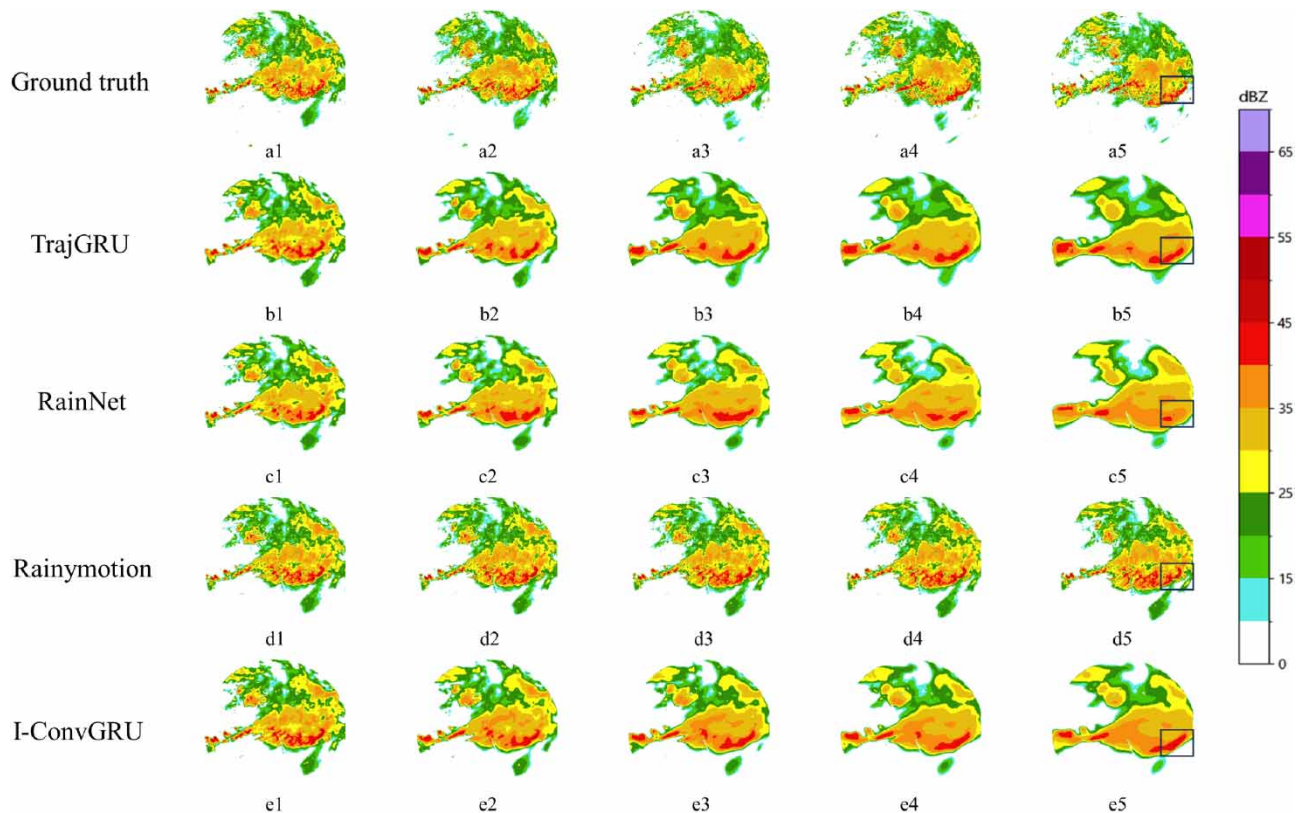


Figure 6 | Visualization results of Case 1, (a1–a5) represent the true ground radar values at 6, 18, 30, 42, and 60 min; (b1–b5), (c1–c5), (d1–d5), and (e1–e5) the corresponding time prediction result diagrams of TrajGRU, RainNet, Rainymotion, and I-ConvGRU models, respectively.

with prediction times of 6, 18, 30, 42, and 60 min. We use a, b, c, d, and e to represent the ground observation image and TrajGRU, respectively. Rainnet, Rainymotion, and the I-ConvGRU models predict images.

Use the numbers 1, 2, 3, 4, and 5 to represent the echo images at different times. It can be seen from Figure 6 that as the prediction time increases, the prediction ability of the prediction images of each model becomes worse and worse, and the loss of details is greater. There are also certain differences from the ground observation images in the echo size and echo boundary. These problems inevitably appear in the results predicted by the ConvLSTM, CSAConvLSTM (Xiong *et al.* 2021), and SmaAt-UNet (Nie *et al.* 2021) models. When the prediction time is 60 min, it can be seen that the I-ConvGRU model performs better. In the strong echo area (black box), it can be seen that although the image of the optical flow method is relatively clear, it overestimates the intensity and range of rainfall, while the convolution-based method, the cumulative RainNet model, underestimates rainfall intensity and extent. However, there is a large deviation between the strong echo position predicted by the TrajGRU model based on the recurrent network and the true position. Only the I-ConvGRU model is basically consistent with the actual echo intensity and echo area.

To analyze the warning ability of the model more intuitively, we present the Taylor diagrams of the different models in Case 1, thus helping to analyze the root mean square deviation, standard deviation, and correlation coefficient between the model simulation results and the observed data. As seen from the plots, the standard deviation of the true value of the 60-min prediction is about 60, indicating that the distribution of precipitation intensity in the radar image is not uniform, thus verifying the effectiveness of the weighted loss function we used for early warning. Additionally, Figure 7 shows that the correlation value of the I-ConvGRU model (red squares) is closest to 1, and the root mean square error is closest to the X-axis observation. Therefore, it can be concluded that the I-ConvGRU model demonstrates the best performance.

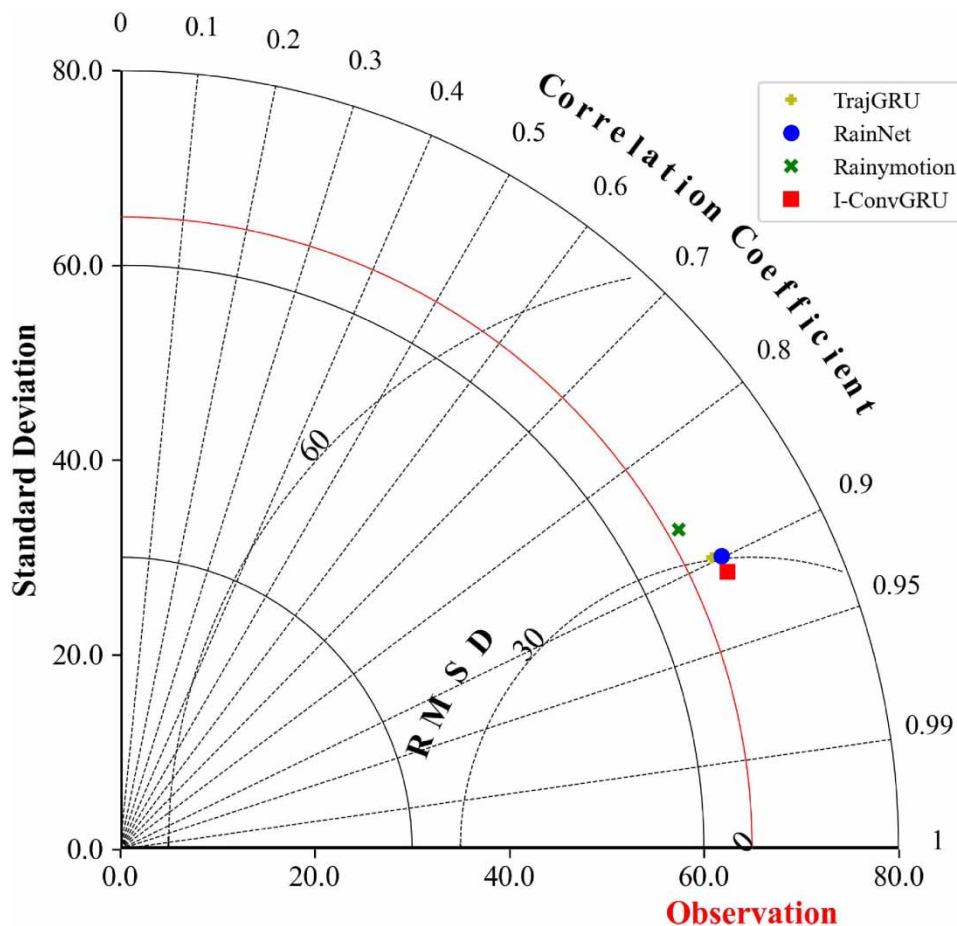


Figure 7 | Taylor diagrams of radar echograms predicted by different models in Case 1.

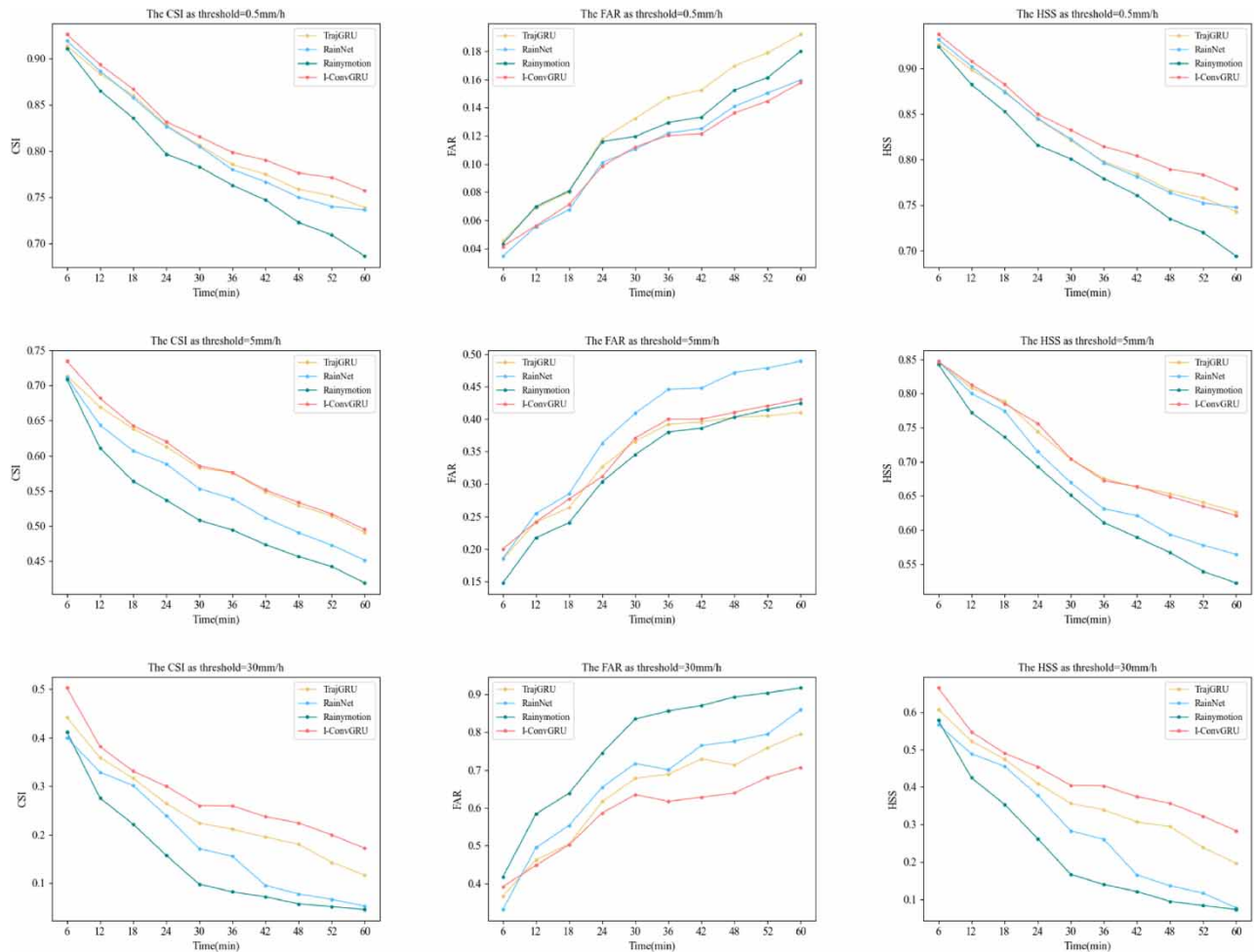


Figure 8 | In Case 1, the CSI, FAR, and HSS values change over time when the thresholds are 0.5, 5, and 30 mm/h.

To specifically quantify the predictive power of the model. We analyzed rainfall's CSI, FAR, and HSS values under three different conditions: precipitation threshold $R \geq 0.5$ mm/h, $R \geq 5$ mm/h, and $R \geq 30$ mm/h. Figure 8 shows the changes in various variables over time under different precipitation thresholds. It can be seen that under different thresholds, as time increases, the CSI and HSS values gradually decrease, and the FAR value gradually increases, corresponding to the test set's results. When the threshold $R \geq 5$ mm/h, the FAR value of the model based on Rainymotion is smaller than that of the I-ConvGRU model, which is again verified with the test set results. Analyzing the overall situation, when the threshold value is larger, the HSS, CSI, and FAR values corresponding to the I-ConvGRU model perform better and better than other models. When the threshold $R \geq 30$ mm/h is fully analyzed, it is found that the longer the time, the better the effect of the model is compared with other models. The model adds new warning pictures and incorporates new information over time. More strong echo features are extracted, making the model perform better when the threshold is larger. In the case of weak precipitation, the FAR value of the model is worse than that of other models. This may be due to the iterative introduction of more erroneous weak precipitation features, gradually increasing the error. However, when comprehensively comparing the model's effectiveness, its CSI and HSS values perform better than other models. So it can be concluded that the performance of the I-ConvGRU model is better than other models.

In order to more effectively determine that the I-ConvGRU model is superior to other models in its warning ability for heavy rainfall events over time. We generated violin plots of radar images for each model to assess early warning performance at 60 min for individual scenarios (Figure 9). Analysis of the Taylor plot indicates that the width of the violin plot for the

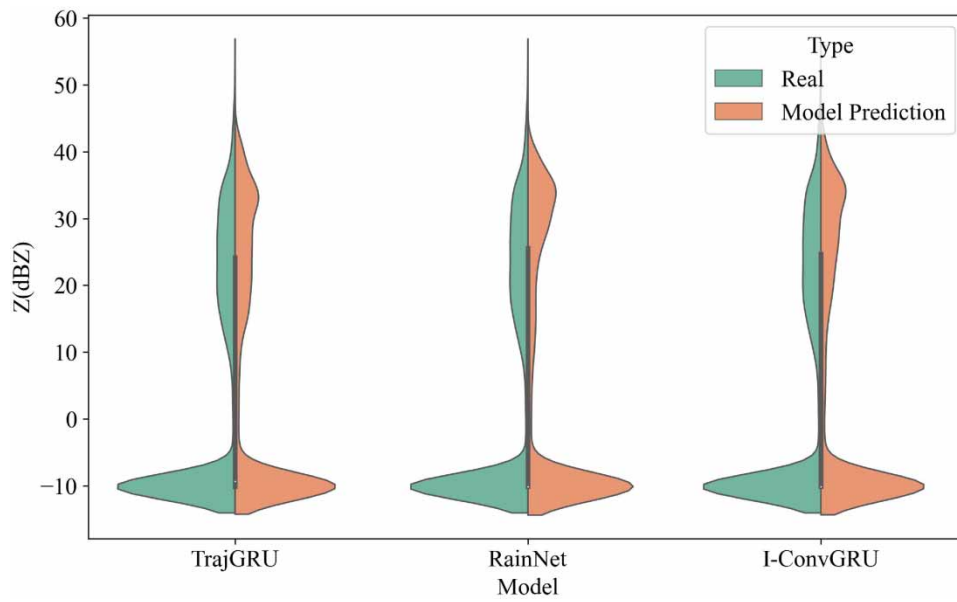


Figure 9 | Violin plots of reflectivity values of radar echo warning maps for different models at a prediction time of 60 min.

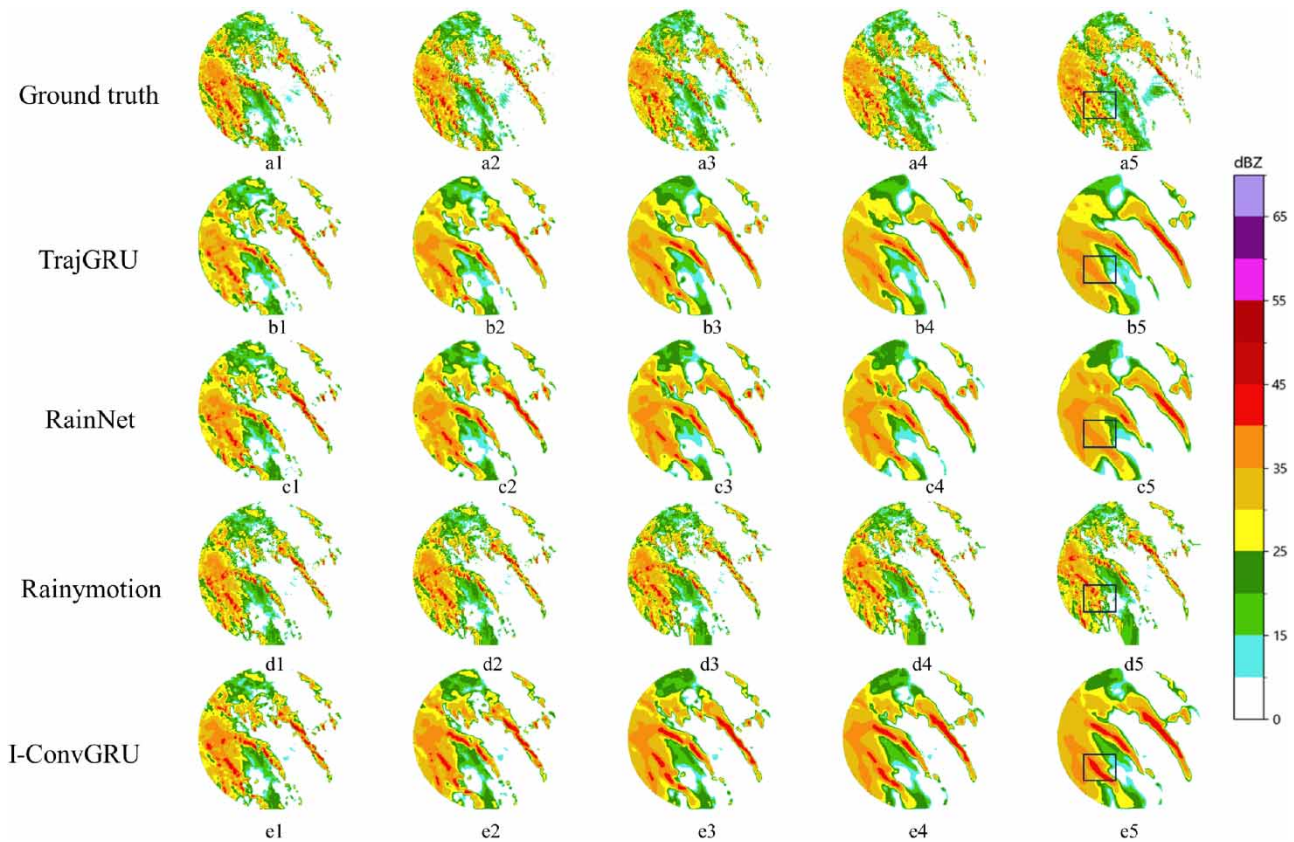


Figure 10 | Visualization results of Case 2, (a1–a5) represent the true ground radar values at 6, 18, 30, 42, and 60 min; (b1–b5), (c1–c5), (d1–d5), and (e1–e5) the corresponding time prediction result diagrams of TrajGRU, RainNet, Rainymotion, and I-ConvGRU models, respectively.

I-ConvGRU model closely aligns with the actual width, given a reflectivity of 40 dBZ over time. This suggests that the I-ConvGRU model exhibits the most robust warning capability.

Case 2

In addition, we randomly selected a case with heavy rainfall. The forecast for the time period 13:00–15:00 noon on October 5, 2015, in Hong Kong was selected. The prediction effect is shown in Figure 10. When the prediction time is 60 min, the TrajGRU model and Rainet model in the black box area in the echo map have poor early warning capabilities in heavy precipitation conditions and do not predict the echo boundary and echo intensity similar to the actual map. Although the optical flow method predicts echo intensities similar to ground observation images, there are certain errors in the echo boundaries and echo ranges it predicts. Only the I-ConvGRU model predicts strong echo areas, similar to the actual image echo boundaries. At the same time, it also once again verified the effectiveness of the I-ConvGRU model in predicting strong echo areas compared with other types of models in long-term prediction.

In this example, we evaluated the I-ConvGRU model's performance by calculating the CSI, HSS, and FAR at various precipitation thresholds and intervals, detailed in Figure 11. At lower thresholds ($R = 0.5$ mm/h and $R = 5$ mm/h), the I-ConvGRU model's performance closely matched that of the TrajGRU model. However, at a higher threshold ($R = 30$ mm/h), it exhibited superior CSI and HSS values and a lower FAR than the RainNet and TrajGRU models. These results

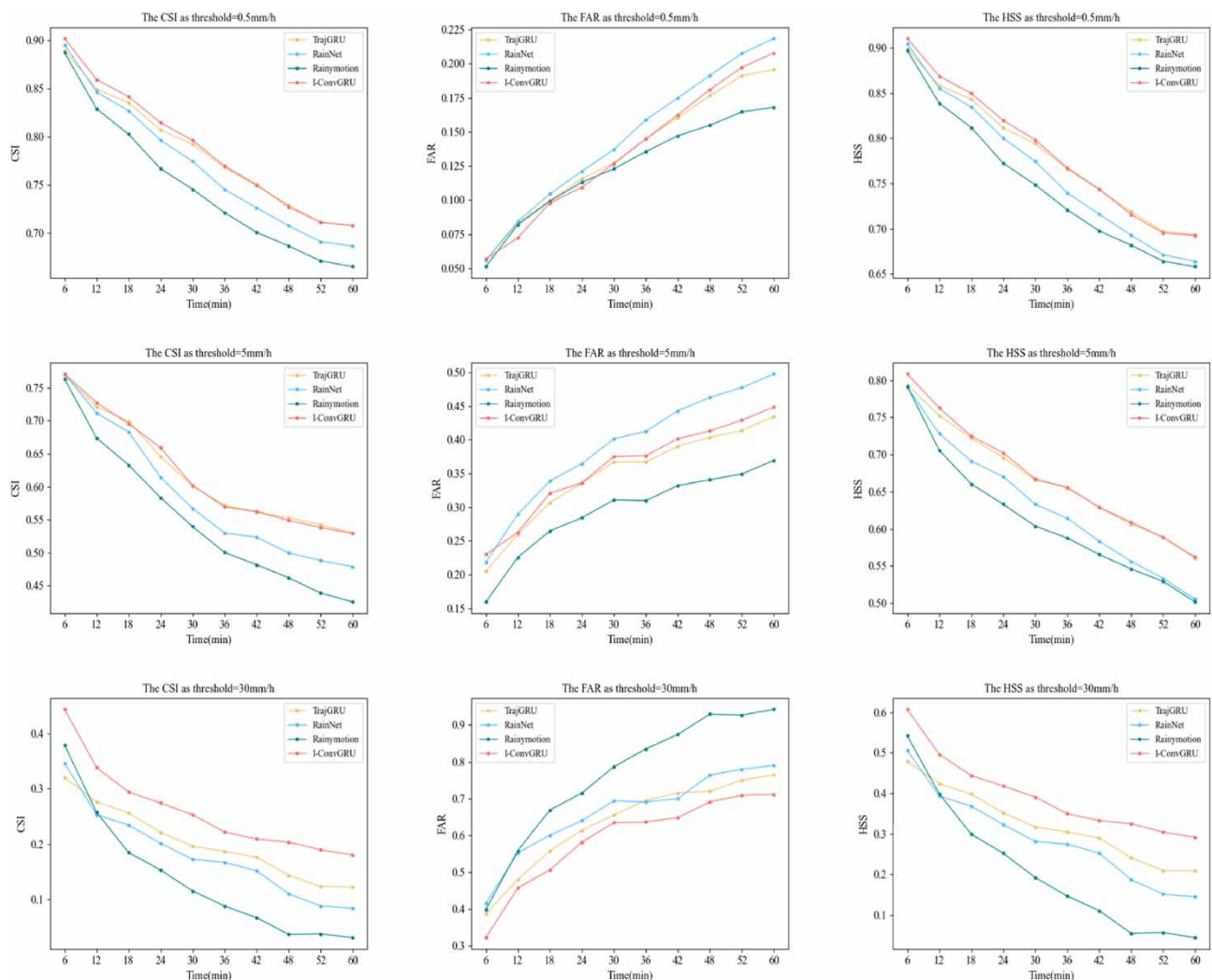


Figure 11 | Changes of CSI value, FAR value, and HSS value over time in Case 2 when the thresholds are 0.5, 5, and 30 mm/h.

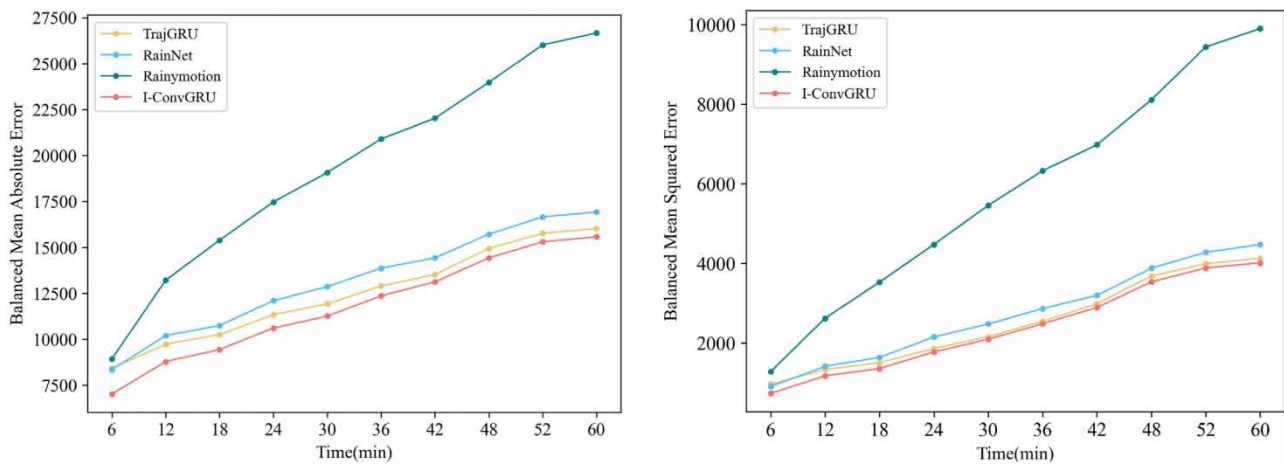


Figure 12 | Plot of B-MAE and B-MSE over time for each model in Case 2.

corroborate previous findings, affirming the I-ConvGRU model's strong predictive capability for heavy rainfall, akin to the performance of the T-Unet model in early warning scenarios of heavy precipitation.

In addition, we present plots illustrating the temporal evolution of B-MSE and B-MAE values for various models in Figure 12. In-depth verification of the radar echo extrapolation warning process is provided through graphs depicting the temporal evolution of B-MSE and B-MAE values for different models. Analysis reveals that Rainymotion exhibits higher B-MSE and B-MAE values, indicating relatively poorer warning capability. Among the remaining three models, the I-ConvGRU model demonstrates relatively smaller B-MSE and B-MAE values, potentially attributable to its incorporation of output signal information, thereby enhancing early warning capability. Among the remaining three models, the B-MSE and B-MAE values of the I-ConvGRU model are relatively small, which may be due to the fact that the I-ConvGRU model incorporates the information of the output signals and improves the early warning capability.

CONCLUSIONS

This study innovatively merges the temporal dynamics of the ConvGRU model with the loop iteration strengths of the RainNet model to forge the I-ConvGRU model. This hybrid approach marries the temporal feature extraction capabilities of the ConvGRU network with the iterative optimization advantages of RainNet, enabling a more granular optimization of model parameters through the iterative input of 10 sequential time-series images. This process iteratively forecasts future images, which are then cycled back into the network after removing the initial input image, thus refining the parameters with each iteration. Traditional encoder-predictor networks primarily rely on circular convolutions and fall short of capturing spatial characteristics. To address this, the new model introduces skip connections after each sampling to enrich the model's spatial feature representation.

Comprehensive analysis yields the following conclusions:

- (1) Visualization of two case studies reveals that the I-ConvGRU model excels in predicting strong echo signals, closely matching the intensity and boundaries of ground-observed echoes. While Rainymotion displays clearer images, its echo boundaries and size predictions significantly deviate from actual observations. This is because the essence of the optical flow method is to predict the movement of pixels. It will not change the size of the pixel value. Similarly, the TrajGRU and RainNet models struggle with accurate echo size predictions. Neither of these two models uses the output signal. Therefore, the echo size feature is insufficient.
- (2) Comparative testing indicates that the I-ConvGRU model achieves a 3.8 and 3.2% reduction in MSE and MAE, respectively, versus the TrajGRU model; it also sees improvements in B-MSE and B-MAE by 5.8 and 3.8%. Moreover, critical meteorological metrics, CSI and HSS, improve by 9.6 and 6.8% for precipitation rates ≥ 30 mm/h compared with the TrajGRU model, showcasing superior early warning performance, especially in heavy rainfall scenarios. This is because the traditional encoder-predictor network makes less use of the output signal and cannot extract enough strong echo features.

However, although the I-ConvGRU model improves the accuracy of precipitation proximity forecasting, it, like other deep learning models, faces the situation that it loses too much detail in the later stages of the forecast and is unable to maintain the resolution as in the optical flow method, which leads to a significant decrease in resolution. This is because the widespread use of MSE or MAE loss functions smooths the prediction results, resulting in blurred extrapolated radar images; and the forecasting ability for light rain is weak, which is also reflected in the RainNet model. And there is no guarantee that the best warning rate is maintained in every threshold case. These drawbacks are inevitable in HPRNN (Jing *et al.* 2020) and ConvLSTM (Shi *et al.* 2015). Finally, the uneven distribution of HKO-7 data set may affect the early warning result to some extent. In future experiments, in order to increase the resolution, we consider introducing GANs and new loss function boosting models to improve the forecasting ability for light rain and consider using data sets of heavy precipitation to improve the prediction performance of the model. Also consider adding information from multiple input variables to the model for prediction, such as speed and temperature. Humidity and other variables because speed can be more specifically related to time and space. Temperature and humidity because storms, rainfall formation are inextricably linked to temperature. Also, physical constraint equations are considered to optimize the warning rate in combination with deep learning. Physical conditions help the model to better understand the physical meaning of the data. Deep learning helps to learn the nonlinear relationships of the data. Combining the two with each other can better capture the patterns and features in the data and improve the prediction accuracy.

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DATA AVAILABILITY STATEMENT

All relevant data are available from an online repository or repositories. The HKO-7 dataset used in this study is from the Hong Kong Observatory at https://github.com/sxjscien-ce/HKO-7/tree/master/hko_data (accessed on 20 November 2021).

CONFLICT OF INTEREST

The authors declare there is no conflict.

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